

- 6 A nichrome wire of cross-sectional area $5.7 \times 10^{-8} \text{ m}^2$ is connected to a power supply. The potential difference (p.d.) across the wire is 230 V. The power dissipated in the wire is 1.0 kW.

(a) Show that the resistance of the wire is 53Ω .

[2]

(b) The resistivity of nichrome is $1.3 \times 10^{-6} \Omega \text{ m}$.

Calculate the length of the wire.

length = m [2]

(c) The average drift speed of the charge carriers (free electrons) in the nichrome wire is $5.4 \times 10^{-3} \text{ m s}^{-1}$.

Calculate the number density of charge carriers of nichrome.

number density = m^{-3} [3]

[Total: 7]